

Ewald sums for a 2D two-component plasma

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1 Math

1.1 Special functions

The exponential integral function is defined as

$$E_1(x) = \int_x^\infty dt \frac{e^{-t}}{t} \rightarrow \begin{cases} \frac{e^{-x}}{x}, & x \rightarrow \infty, \\ -\ln x - \gamma, & x \rightarrow 0, \end{cases} \quad (1)$$

where γ is Euler's constant.

The error function is defined

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dt e^{-t^2}. \quad (2)$$

The complementary error function is

$$\operatorname{erfc}(x) = 1 - \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty dt e^{-t^2} = \frac{2}{\sqrt{\pi}} \int_0^{1/x} dt \frac{e^{-1/t^2}}{t^2}. \quad (3)$$

Useful limiting behaviors are

$$\operatorname{erf}(x) \rightarrow \begin{cases} 1 - \frac{e^{-x^2}}{\sqrt{\pi}x}, & (x \rightarrow \infty), \\ \frac{2x}{\sqrt{\pi}}, & (x \rightarrow 0). \end{cases} \quad (4)$$

1.2 Lattice things

Consider a 2D Bravais lattice defined by primitive vectors $\{\mathbf{a}_1, \mathbf{a}_2\}$. A general lattice vector is

$$\mathbf{R} = l_1 \mathbf{a}_1 + l_2 \mathbf{a}_2, \quad l_i \in \mathbb{Z}. \quad (5)$$

The primitive unit-cell area is $\Omega = |\mathbf{a}_1 \times \mathbf{a}_2|$. The reciprocal lattice is defined by $\{\mathbf{b}_1, \mathbf{b}_2\}$ such that $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij}$. A general reciprocal-lattice vector is

$$\mathbf{G} = l_1 \mathbf{b}_1 + l_2 \mathbf{b}_2, \quad l_i \in \mathbb{Z}. \quad (6)$$

A useful mathematical identity is

$$\sum_{\mathbf{G}} e^{i\mathbf{G}\cdot\mathbf{x}} = \Omega \sum_{\mathbf{R}} \delta(\mathbf{x} - \mathbf{R}). \quad (7)$$

This may be used to derive the “2D theta-transform” identity:

$$\sum_{\mathbf{R}} e^{-|\mathbf{r}-\mathbf{R}|^2\xi^2} = \frac{\pi}{\Omega\xi^2} \sum_{\mathbf{G}} e^{-G^2/4\xi^2} e^{-i\mathbf{G}\cdot\mathbf{r}}. \quad (8)$$

Proof:

$$\sum_{\mathbf{R}} e^{-|\mathbf{r}-\mathbf{R}|^2\xi^2} = \int d^2\mathbf{r}' e^{-r'^2\xi^2} \sum_{\mathbf{R}} \delta(\mathbf{r}' - \mathbf{r} - \mathbf{R}) \quad (9)$$

$$= \frac{1}{\Omega} \sum_{\mathbf{G}} \int d^2\mathbf{r}' e^{-r'^2\xi^2} e^{i\mathbf{G}\cdot(\mathbf{r}'-\mathbf{r})} \quad (10)$$

$$= \frac{1}{\Omega} \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} \int d^2\mathbf{r}' e^{-r'^2\xi^2} e^{i\mathbf{G}\cdot\mathbf{r}'} \quad (11)$$

$$= \frac{\pi}{\Omega\xi^2} \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{r}} e^{-G^2/4\xi^2}. \quad \square \quad (12)$$

2 Coulomb energy in a periodic system

2.1 Setup and interaction potential

Consider a system of N charged particles q_i with coordinates $\{\mathbf{r}_1, \dots, \mathbf{r}_N\}$ confined to a simulation cell, satisfying charge neutrality $\sum_i q_i = 0$. We impose periodic boundary conditions, which means our simulation cell is part of a larger “super-lattice.” We take our fundamental cell to be defined by primitive vectors $\{\mathbf{a}_1, \mathbf{a}_2\}$ and the super-lattice is spanned by $\mathbf{R} = l_1\mathbf{a}_1 + l_2\mathbf{a}_2$. The area of the primitive cell is $\Omega = |\mathbf{a}_1 \times \mathbf{a}_2|$.

We are ultimately interested in the total energy of charges q and q' interacting with the logarithmic potential $v(\mathbf{r}) \sim -qq' \ln r$. In a macroscopic system, however, this interaction requires regularization. We regularize it by using the more general interaction potential between vortices in 2D; see, e.g., [1]:

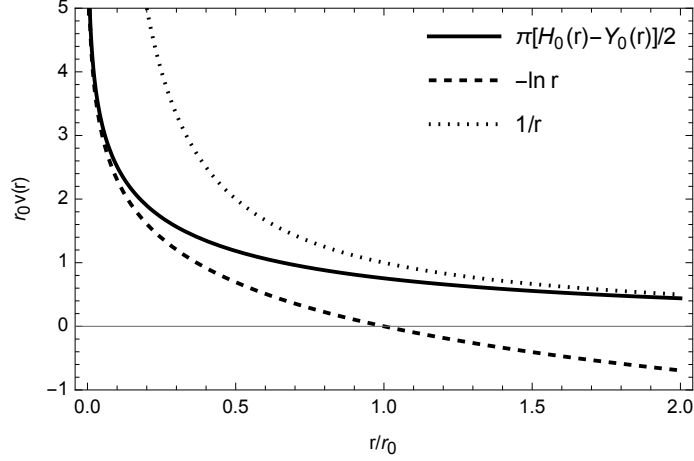
$$v(\mathbf{r}) = \frac{\pi}{2r_0} \left[\mathbf{H}_0\left(\frac{r}{r_0}\right) - Y\left(\frac{r}{r_0}\right) \right] \quad (13)$$

where $\mathbf{H}_n(z)$ and $Y_n(z)$ are the Struve function and modified Bessel function of the second kind, respectively. The Fourier transform of the interaction is

$$v(\mathbf{k}) = \frac{1}{1 + r_0 k} \frac{2\pi}{k}. \quad (14)$$

The interaction has limiting behaviors

$$v(\mathbf{r}) \rightarrow \begin{cases} -\frac{1}{r_0} \ln\left(\frac{r}{r_0}\right), & r_0 \rightarrow \infty \\ \frac{1}{r}, & r_0 \rightarrow 0 \end{cases}. \quad (15)$$



At the end of our calculations, we will take $r_0 \rightarrow \infty$ to recover the desired logarithmic interaction.

Note that, in semiconductor physics, the interaction (13) is known as the “Rytova-Keldysh” potential, where it also describes the interaction between electrons in a 2D layer with intrinsic susceptibility $\chi_0 = r_0/2\pi$.

2.2 Ewald splitting

To implement the Ewald splitting technique for the potential energy sum, we write

$$v(\mathbf{r}) = \int_0^\infty dt e^{-tr^2/r_0^2} \Phi(t), \quad \Phi(t) = \frac{1}{r_0} \frac{e^{1/4t}}{2t} \operatorname{erfc}\left(\frac{1}{2\sqrt{t}}\right). \quad (16)$$

This identity can be checked explicitly in `Mathematica`.

We then split $v(\mathbf{r}) = v_{\text{SR}}(\mathbf{r}) + v_{\text{LR}}(\mathbf{r})$, where SR and LR refer to “short range” and “long range” respectively:

$$v_{\text{SR}}(\mathbf{r}) = \int_{t_0}^\infty dt e^{-tr^2/r_0^2} \Phi(t), \quad v_{\text{LR}}(\mathbf{r}) = \int_0^{t_0} dt e^{-tr^2/r_0^2} \Phi(t). \quad (17)$$

We would like to first compute the interaction energy between a pair of charge q_i and q_j (assuming $i \neq j$):

$$E_{\text{pair}}(\mathbf{r}_i - \mathbf{r}_j) = q_i q_j \sum_{\mathbf{R}} v(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}) \quad (18)$$

$$= q_i q_j \sum_{\mathbf{R}} v_{\text{SR}}(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}) + q_i q_j \sum_{\mathbf{R}} v_{\text{LR}}(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}). \quad (19)$$

In the short-range part, we can immediately take $r_0 \rightarrow \infty$ to obtain

$$v_{\text{SR}}(\mathbf{r}) = \frac{1}{2r_0} E_1(t_0 r^2/r_0^2), \quad (r_0 \rightarrow \infty), \quad (20)$$

where $E_1(z)$ is the exponential integral function defined in (1).

For the long-range part, we use Eq. (8) to write

$$\sum_{\mathbf{R}} v_{\text{LR}}(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}) = \int_0^\infty dt \Phi(t) \sum_{\mathbf{R}} e^{-t(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R})^2 / r_0^2} \quad (21)$$

$$= \frac{\pi r_0^2}{\Omega} \int_0^\infty dt \frac{\Phi(t)}{t} \sum_{\mathbf{G}} e^{-G^2 r_0^2 / 4t} e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \quad (22)$$

$$= \frac{\pi r_0^2}{\Omega} \sum_{\mathbf{G}} e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)} \int_0^\infty dt \frac{\Phi(t)}{t} e^{-G^2 r_0^2 / 4t}. \quad (23)$$

At this stage, we may safely take $r_0 \rightarrow \infty$ in $\Phi(t)$ to obtain

$$\sum_{\mathbf{R}} v_{\text{LR}}(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}) = \frac{2\pi}{\Omega r_0} \sum_{\mathbf{G}} \frac{e^{-G^2 r_0^2 / 4t_0}}{G^2} e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)}, \quad (r_0 \rightarrow \infty). \quad (24)$$

The divergent $\mathbf{G} = 0$ term should be separated out; its contribution will vanish when summing over charges and using the charge neutrality condition. We have

$$\begin{aligned} \sum_{\mathbf{R}} v_{\text{LR}}(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R}) &= \frac{2\pi}{\Omega r_0} \sum_{\mathbf{G} \neq 0} \frac{e^{-G^2 r_0^2 / 4t_0}}{G^2} e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)} - \frac{\pi}{2\Omega r_0} (\mathbf{r}_i - \mathbf{r}_j)^2 \\ &\quad + \frac{2\pi}{\Omega r_0 G^2} \Big|_{G=0} - \frac{\pi r_0}{2\Omega t_0}. \end{aligned} \quad (25)$$

To arrive at this result, we have done an angular average in the term $\sim (\mathbf{r}_i - \mathbf{r}_j)^2$. I think is sensible if we want the result to respect the isotropy of the system in the thermodynamic limit.

We finally obtain

$$\begin{aligned} E_{\text{pair}}(\mathbf{r}_i - \mathbf{r}_j) &= \frac{q_i q_j}{2r_0} \sum_{\mathbf{R}} E_1 \left(\frac{t_0 (\mathbf{r}_i - \mathbf{r}_j - \mathbf{R})^2}{r_0^2} \right) \\ &\quad + \frac{2\pi q_i q_j}{\Omega r_0} \sum_{\mathbf{G} \neq 0} \frac{e^{-G^2 r_0^2 / 4t_0}}{G^2} e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)} - \frac{\pi q_i q_j}{2\Omega r_0} (\mathbf{r}_i - \mathbf{r}_j)^2 \\ &\quad + q_i q_j \left(\frac{2\pi q_i q_j}{\Omega r_0 G^2} \Big|_{G=0} - \frac{\pi r_0}{2\Omega t_0} \right). \end{aligned} \quad (26)$$

To include the self-energy of a particle interacting with its own periodic images, we consider the limit

$$E_{\text{self}}^i = \lim_{\mathbf{r} \rightarrow 0} \left[E_{\text{pair}}(\mathbf{r}) + \frac{q_i^2}{r_0} \ln \left(\frac{r}{r_0} \right) \right], \quad (27)$$

so that

$$E_{\text{self}}^i = -\frac{q_i^2}{2r_0} [\gamma + \ln(4t_0)] + \frac{q_i^2}{2r_0} \sum_{\mathbf{R} \neq 0} E_1\left(\frac{t_0 R^2}{r_0^2}\right) + \frac{2\pi q_i q_j}{\Omega r_0} \sum_{\mathbf{G} \neq 0} \frac{e^{-G^2 r_0^2 / 4t_0}}{G^2} + q_i^2 \left(\frac{2\pi}{\Omega r_0 G^2} \Big|_{G=0} - \frac{\pi r_0}{2\Omega t_0} \right). \quad (28)$$

The total energy of the system (per unit cell), which is a function of the coordinate of all N particles, is then

$$E[\{\mathbf{r}_i\}] = \sum_{i < j}^N E_{\text{pair}}(\mathbf{r}_{ij}) + \frac{1}{2} \sum_{i=1}^N E_{\text{self}}^i. \quad (29)$$

Using our results above together with the charge neutrality condition, we find

$$E[\{\mathbf{r}_i\}] = -\frac{1}{4}(\gamma + \ln t_0) \sum_{i=1}^N q_i^2 + \frac{1}{4r_0} \sum_{\mathbf{R}}' \sum_{i,j=1}^N q_i q_j E_1\left(\frac{t_0(\mathbf{r}_i - \mathbf{r}_j - \mathbf{R})^2}{r_0^2}\right) + \frac{\pi}{\Omega r_0} \sum_{\mathbf{G} \neq 0} \frac{e^{-G^2 r_0^2 / 4t_0}}{G^2} \sum_{i,j=1}^N q_i q_j e^{-i\mathbf{G} \cdot (\mathbf{r}_i - \mathbf{r}_j)} - \frac{\pi}{4\Omega r_0} \sum_{i,j=1}^N q_i q_j (\mathbf{r}_i - \mathbf{r}_j)^2 \quad (30)$$

where the primed sum on $\mathbf{R}$ mean to exclude the term $\mathbf{R} = 0$ when $i = j$.

At this stage, to facilitate comparison with other references, we make the following changes of notation and simplifications: We choose a square Bravais super-lattice, with lattice spacing a , so that $\mathbf{R} = a\mathbf{n}$, where $\mathbf{n} = n_x \hat{\mathbf{x}} + n_y \hat{\mathbf{y}}$, $\Omega = a^2$, and $\mathbf{G} = 2\pi\mathbf{n}/a$. We then measure all lengths in units a . We also renormalize our unit of charge $q \rightarrow q\sqrt{r_0}$ and finally set $t_0 = \alpha^2 r_0^2$. This way we obtain

$$E[\{\mathbf{r}_i\}] = -\frac{1}{4}(\gamma + \ln \alpha^2) \sum_{i=1}^N q_i^2 + \frac{1}{4} \sum_{\mathbf{n}}' \sum_{i,j=1}^N q_i q_j E_1(\alpha^2(\mathbf{r}_i - \mathbf{r}_j - \mathbf{n})^2) + \frac{1}{4\pi} \sum_{\mathbf{n} \neq 0} \frac{e^{-\pi^2 n^2 / \alpha^2}}{n^2} \left| \sum_{j=1}^N q_j e^{2\pi i \mathbf{n} \cdot \mathbf{r}_j} \right|^2 + \frac{\pi}{2} \left(\sum_{i=1}^N q_i \mathbf{r}_i \right)^2. \quad (31)$$

We may choose $\alpha = \sqrt{\pi}$, so that the two sums have comparable rates of convergence. This result agrees with Ref. [2].

References

- [1] E. H. Brandt, Phys. Rev. B **79**, 134526 (2009).
- [2] J. W. Perram and S. W. de Leeuw, Physica A **109**, 237 (1981).